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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester M.C.A. (CBCS) Degree Examinations
January 2024

22MCA301 – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	What is Artificial Intelligence? Briefly discuss the goals of Artificial Intelligence.	10	L*1	1	1
	b)	Describe the types of Artificial Intelligence.	10	L2	1	1
2.	a)	Discuss in detail the applications of Artificial Intelligence.	10	L1	1	1
	b)	Briefly discuss the Advantages of Artificial Intelligence.	10	L2	1	1
		Unit – II				
3.	a)	Discuss briefly the challenges in developing applications that use Machine Learning.	10	L1	2	1
	b)	How are Machine Learning Techniques classified? Explain with examples.	10	L2	2	1
4.	a)	Discuss the features of Machine Learning. How is Machine Learning different from traditional programming? Explain.	10	L2	2	1
	b)	What is Data Preprocessing? Explain the Data Preprocessing Techniques.	10	L2	2	1
		Unit – III				
5.	a)	Illustrate with an example the use of CART method in Decision tree algorithm.	10	L3	3	1
	b)	Briefly describe Polynomial regression algorithm and its use.	10	L2	3	1
6.	a)	What is the use of Logistic Regression? Briefly discuss how Logistic Regression works.	10	L2	3	1
	b)	Briefly describe Linear Support Vector Machine. Illustrate with an example.	10	L2	3	1
		Unit – IV				
7.	a)	Illustrate with an example how K Means clustering algorithm works.	10	L2	4	1
	b)	Write a note on Principal Component Analysis.	10	L2	4	1
8.	a)	Briefly discuss KNN algorithm. Illustrate with an example.	10	L2	4	1
	b)	What is reinforcement learning? Discuss its types. How is it different from supervised learning?	10	L2	4	1
		Unit – V				
9.	a)	What is Deep Learning? Discuss how a neural network works.	10	L2	5	1
	b)	Describe Gradient Descent.	10	L2	5	1
10.	a)	Briefly discuss the concept of Back Propagation.	10	L2	5	1
	b)	Describe the architecture of Convolutional Neural Network.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
